

Pinecone — Vector Database

What is Pinecone

Pinecone is a fully managed cloud-native vector database — you store, index, and search vectors without managing any infrastructure.

	ChromaDB	Pinecone	FAISS
Hosting	Self-hosted / local	Fully managed cloud	Local library
Setup	Minimal	API key only	Import and use
Scale	Small-medium	Production scale	Research / benchmarking
Cost	Free	Paid (free tier available)	Free
DevOps	None	None	None

Core Concepts

- **Index** — the container where all vectors are stored, similar to a table in a relational DB
- **Namespace** — logical partition within an index to separate data (e.g. by user, tenant, or domain)
- **Upsert** — insert or update vectors by ID
- **Query** — search for top-k nearest vectors to a given query vector
- **Delete** — remove vectors by ID
- **Metadata filtering** — attach key-value metadata to vectors and filter results at query time (e.g. `{"source": "pdf", "year": 2024}`)

LangChain Integration

Swapping ChromaDB for Pinecone in a RAG pipeline is a one-line change:

```
# ChromaDB (local)
from langchain_community.vectorstores import Chroma
vectorstore = Chroma.from_documents(docs, embedding)

# Pinecone (managed)
from langchain_pinecone import PineconeVectorStore
vectorstore = PineconeVectorStore.from_documents(docs, embedding, index_name="your-index")

# Retriever is identical in both cases
retriever = vectorstore.as_retriever()
```

When to Use

Use Pinecone when you need production-scale vector search without managing infrastructure — ideal for RAG pipelines serving real users at scale.

ChromaDB vs Pinecone vs FAISS — When to Use

Scenario	Use
Rapid prototyping, local dev, learning	ChromaDB
Production RAG serving real users	Pinecone
Research, benchmarking, custom pipelines	FAISS
No internet, fully offline	FAISS or ChromaDB
Multi-tenant SaaS with user isolation	Pinecone (namespaces)
Already in LangChain RAG pipeline	ChromaDB (easy swap to Pinecone later)
Massive dataset, GPU acceleration needed	FAISS
Zero DevOps, managed search at scale	Pinecone